



BHS AI Builder Hackathon

Build something real. Ship it today. Text it to your family tonight.

KICKOFF

Why Right Now Is Different

One year ago, building real software meant learning to code. **That changed.**

Old world

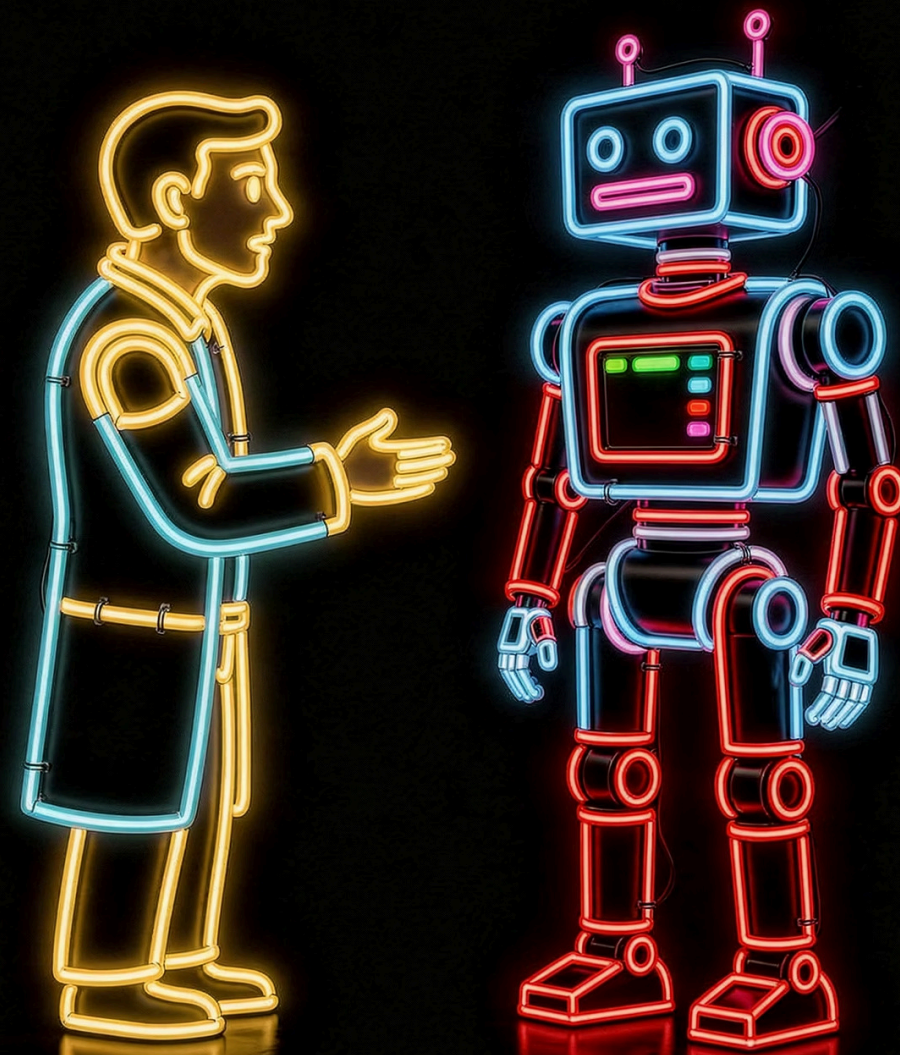
Learn syntax. Debug for hours.
Hope it compiles.

New world

Think clearly. Direct an AI well.
Ship a working app.

The skill that matters

Not memorizing code — **communicating precisely.**



What You're Leaving With

1

Working App

Live at a real URL you can share

1

GitHub Repo

Your own code online, forever

∞

API Credits

Keep building after today

✓ You will ship something you're proud of. That's the only goal.

The AI Coding Tool Landscape

There are a lot of options. Here's the honest one-line breakdown — and why we picked **Claude Code**.

Tool	What it's great at
Claude Code ★	Full agentic coding in your terminal; best MCP ecosystem; beginner-friendly desktop app
Cursor	AI-first code editor; great for experienced devs who want autocomplete on steroids
GitHub Copilot	Inline suggestions as you type; lives inside VS Code
Lovable	Beautiful UIs from plain English; drag-and-drop friendly
Bolt	Fast full-stack prototypes right in the browser
Base44	No-code app builder with a clean visual interface
Replit Agent	Instant cloud environment; zero local setup required

Your Stack Today — Plain English



Claude Code

The AI agent you talk to. It writes, edits, and runs your code.



React

The screens you see — your app's front face (the dining room).



Express + SQLite

Where your data lives. One file = your entire database (the kitchen).



Git + GitHub

Save points and sharing. Your work, safe online, on your own branch.



npm run dev

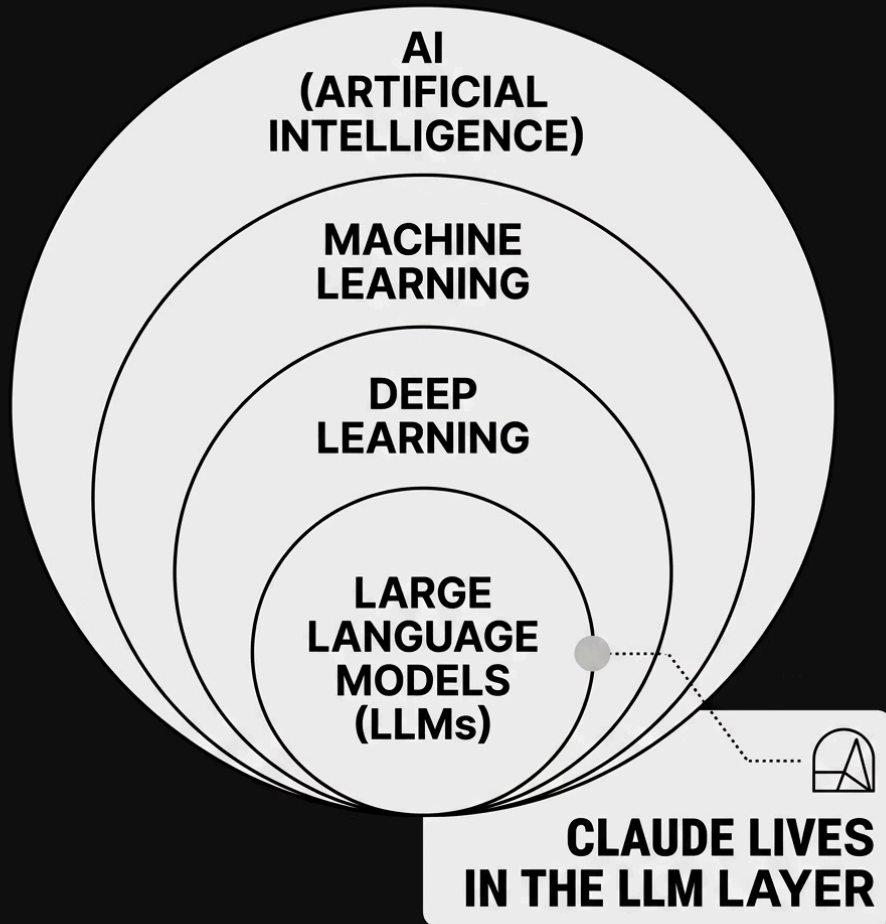
One command. Starts everything. That's it.



Context7

Fetches up-to-date docs so Claude always has the latest info.

AI Is a Slice of a Bigger Field



The Sports Analogy

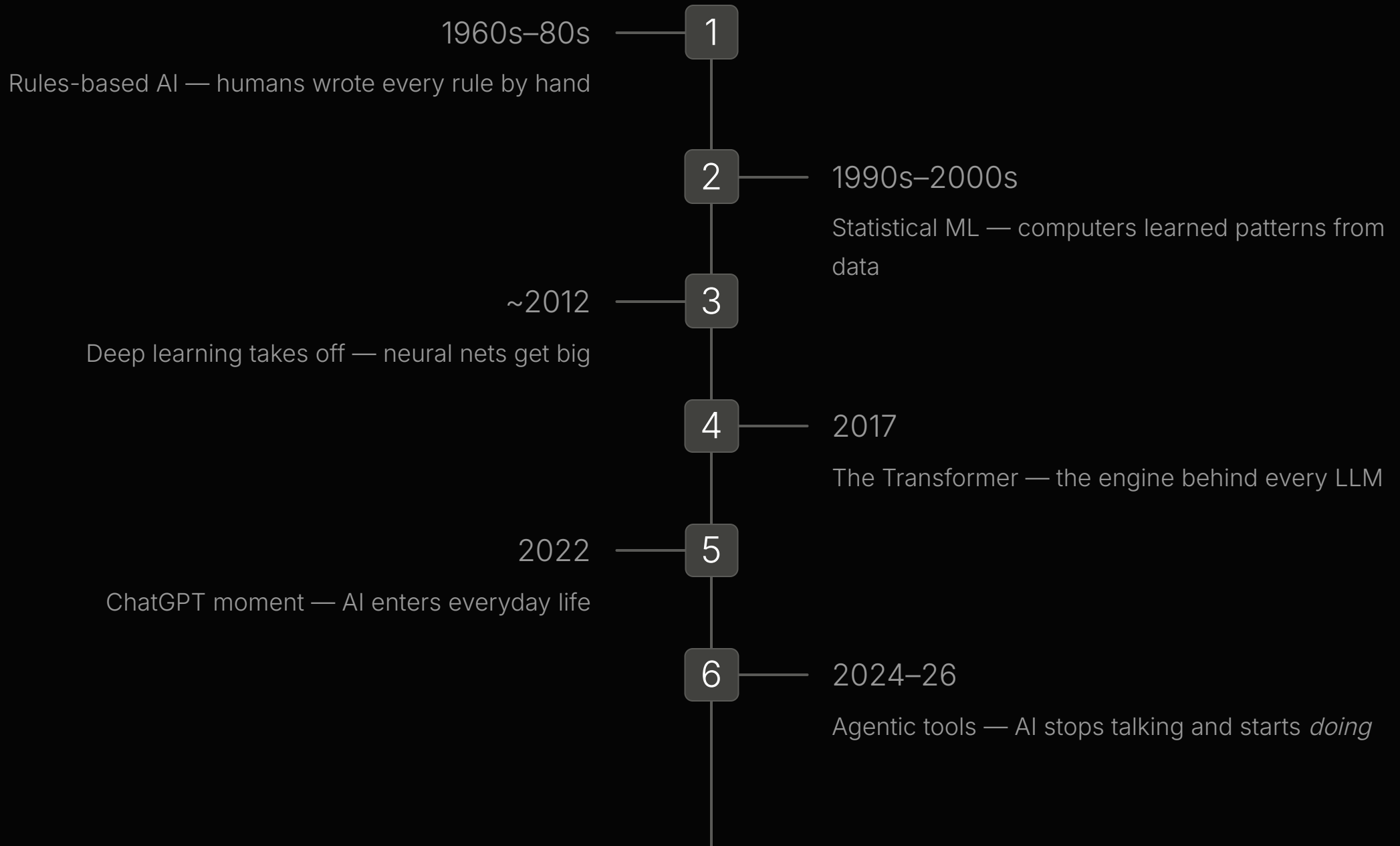
AI = all of sports

Machine Learning = ball sports

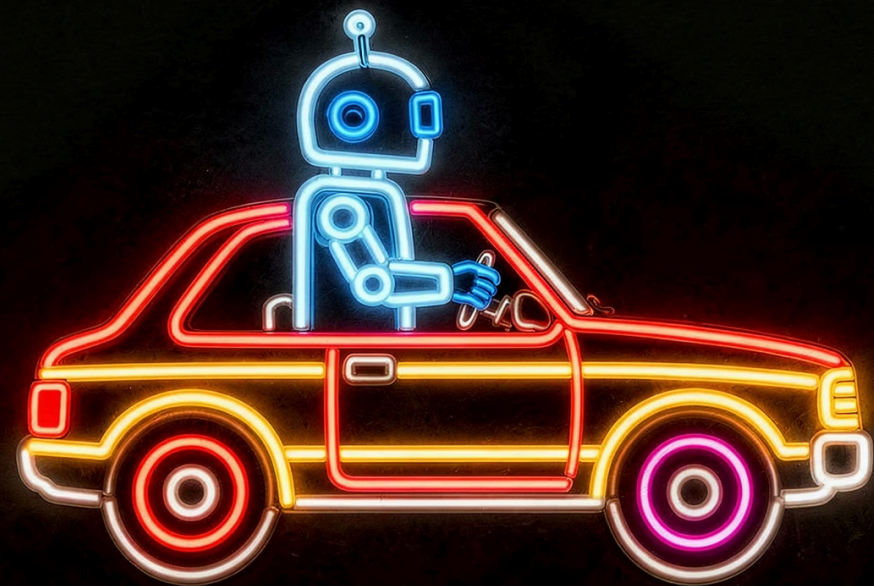
LLMs = basketball — dominant right now, but not the whole game.

- Claude is an LLM. Powerful — but one tool in a much bigger toolbox.

A Quick History of AI



You are building at the most interesting moment in this entire timeline.



Chatbot vs. Agent

Chatbot

You ask. It answers. Done. That's the whole loop.

Agent (Claude Code)

It uses tools, runs commands, edits files, checks results — and **loops until the job is done**. You supervise.

 A chatbot gives you directions. An agent gets in the car and drives.

The AI Vampire Effect

We once believed AI would 100x our output, gifting us abundant free time. The reality is quite different.

Feeling endlessly productive, we find it harder to switch off. We work **more**, caught in a cycle of continuous creation, always pushing for another task.

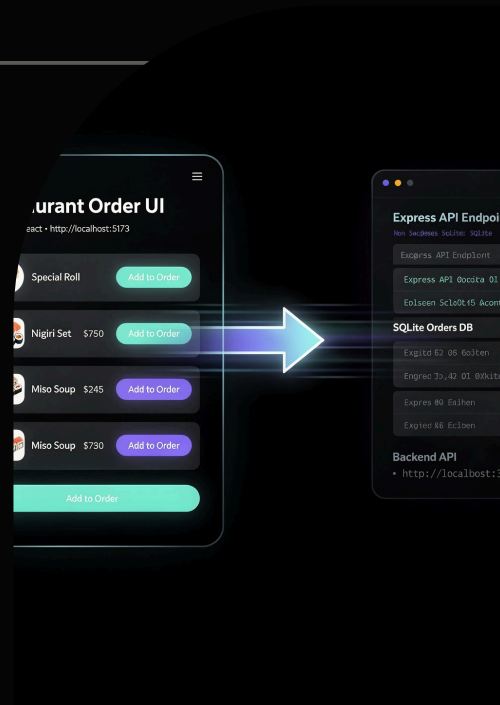
This phenomenon is known as being an **AI Vampire** — forever drawn to the glowing screen of infinite output.



Frontend vs. Backend

Frontend

The dining room — what you see and click.



Backend

The kitchen — where data is stored.

The Restaurant Analogy

- **Frontend** = the dining room and menu — buttons, screens, colors
- **Backend** = the kitchen and fridge — data, logic, storage
- **API** = the order window — the agreed way they talk to each other

📌 Your customers never see the kitchen. They just want the food.

What Is an API?

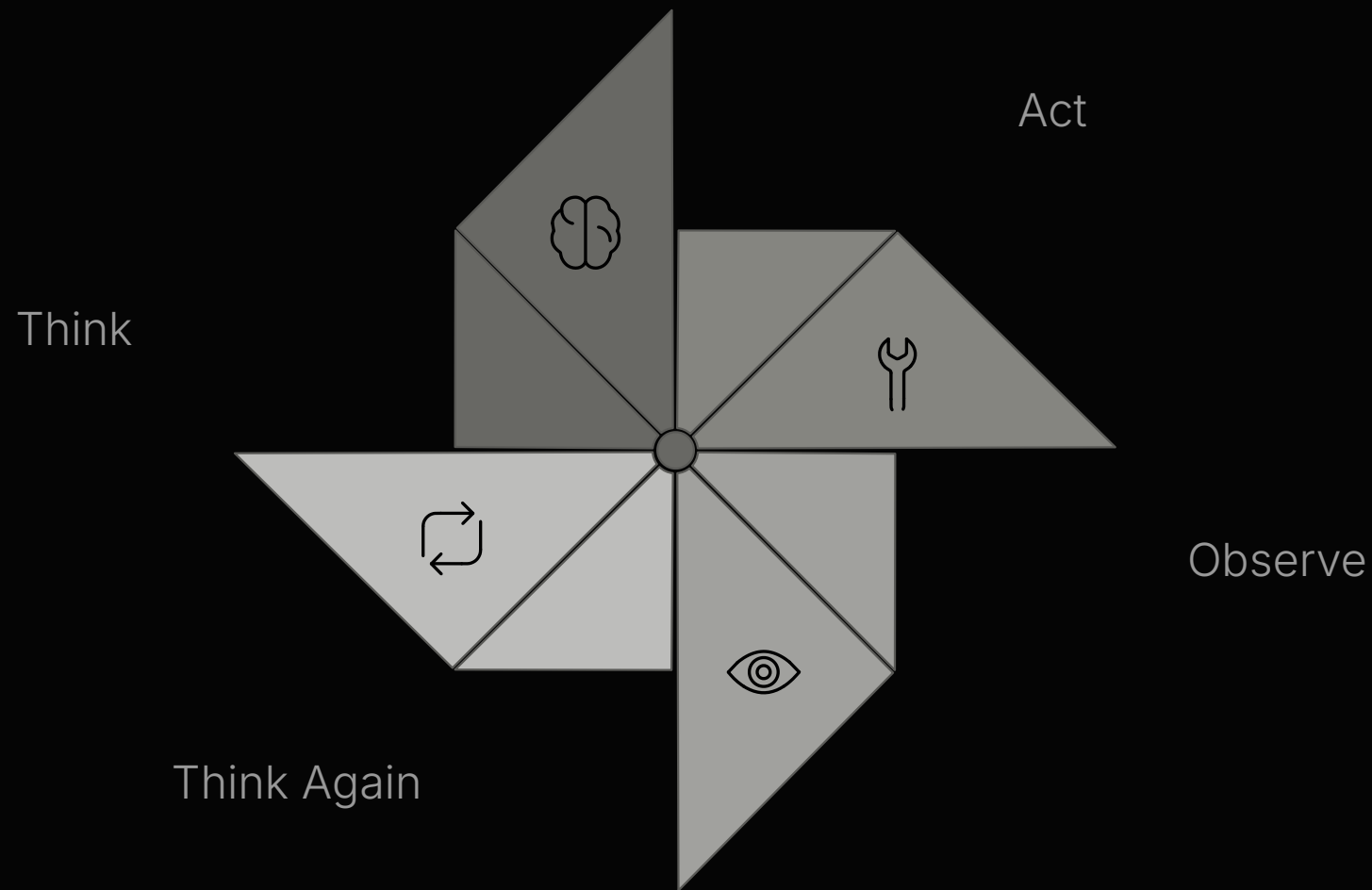


A doorway between programs

An API is an agreed set of rules for how one program asks another for something — and what it gets back.

- You don't walk into the kitchen — you use **the window**
- Order in the expected format → food comes back
- Your frontend talks to your backend through an API
- When your app uses Claude, it goes through **Claude's API**

How Claude Code Actually Works



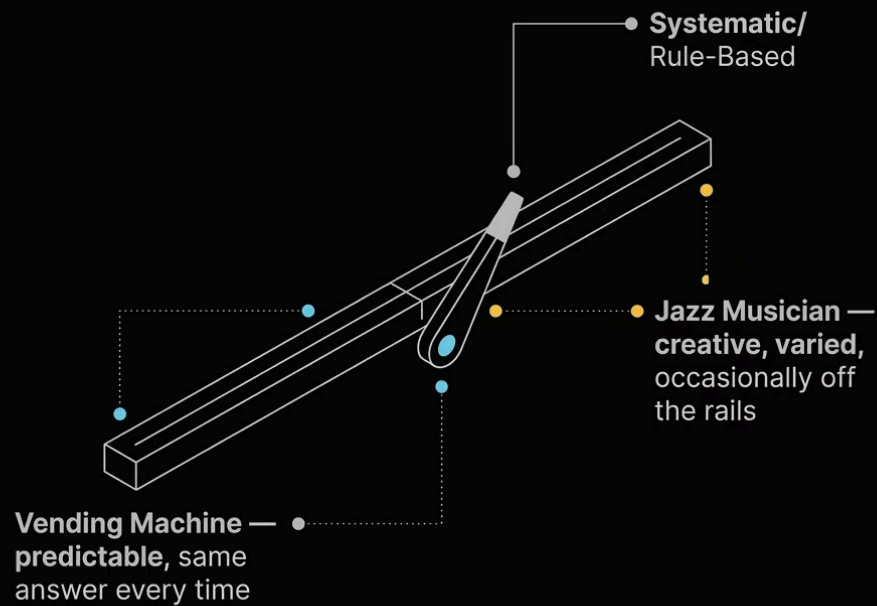
The model is a brilliant chef. The orchestrator is the kitchen manager — handing it ingredients, tasting each dish, and sending it back until it's right.

Temperature: Same Prompt, Different Answers

Type the *exact same prompt* twice. Get two different answers. That's not a bug — it's a dial you control.



Temperature Is a Dial

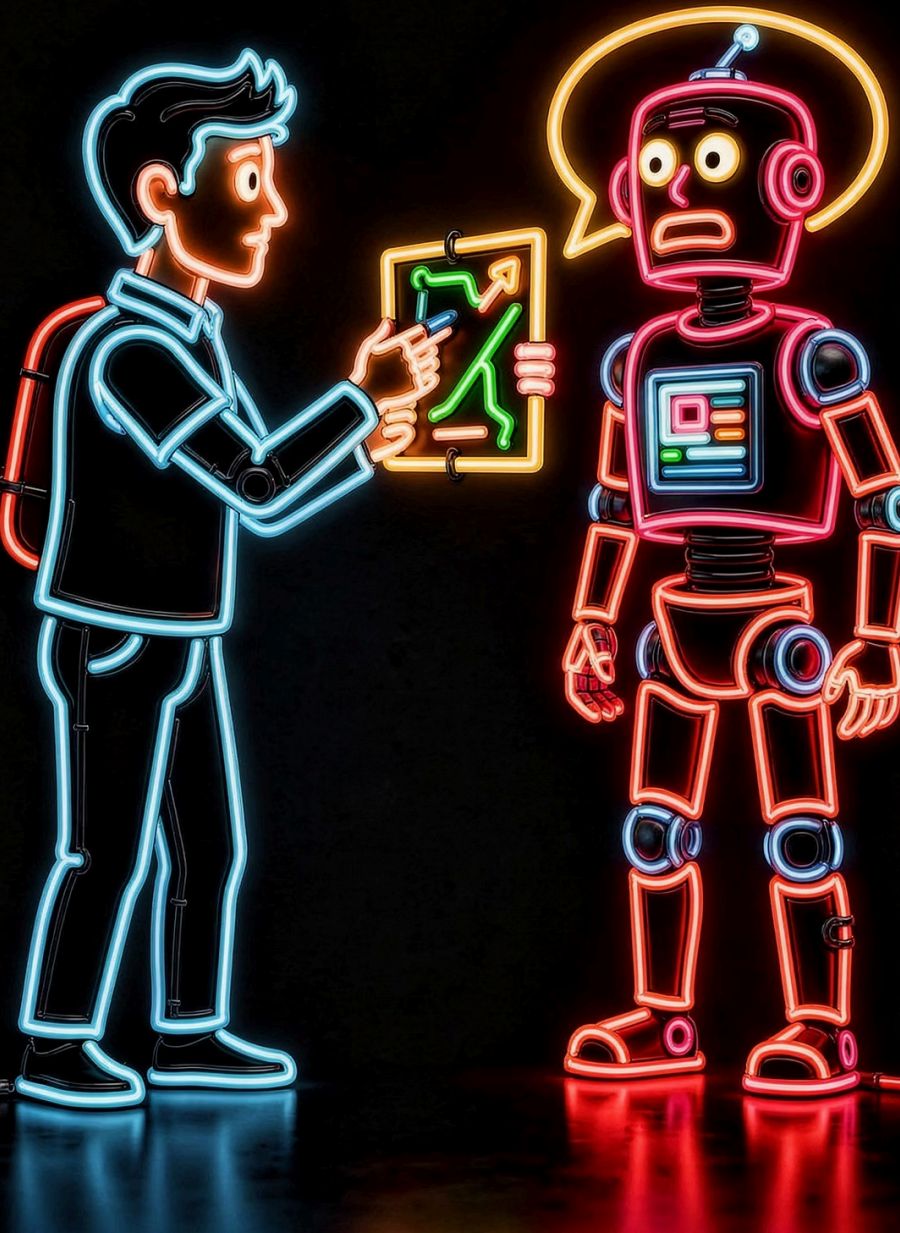


Turn it DOWN when you need:

- Code that runs correctly
- Facts pulled from a document
- A rules-following help bot

Turn it UP when you want:

- Brainstormed names or taglines
- Song lyrics or story ideas
- Varied game-character dialogue



AI Does What You Say — Not What You Mean

Vague prompt

"Make it look better" → the AI guesses. Usually wrong.

Specific prompt

"Change the button color to yellow, increase font size to 18px, and add padding of 12px" → done exactly.



The AI is not lazy — it's **literal**. Be specific and directed.

Plan Before You Prompt



30 seconds of thinking beats 10 chatty prompts.

Before asking the AI to build, describe it in 3–4 sentences:

- What does it **do**?
- Who is it **for**?
- What should it **look like**?
- Is it small enough to **finish today**?

❏ You wouldn't tell a contractor "build me a house" and walk away. Describe the rooms first.

Skills + Integrations = Superpowers

Giving Claude extra tools turns a good result into a great one. These are called **MCP tools** — and they let the AI reach outside itself.



Context7

Fetches live docs so Claude never uses outdated code



Web Search

Looks up real-time info your model wasn't trained on



File System

Reads and writes files directly in your project



Shell / Terminal

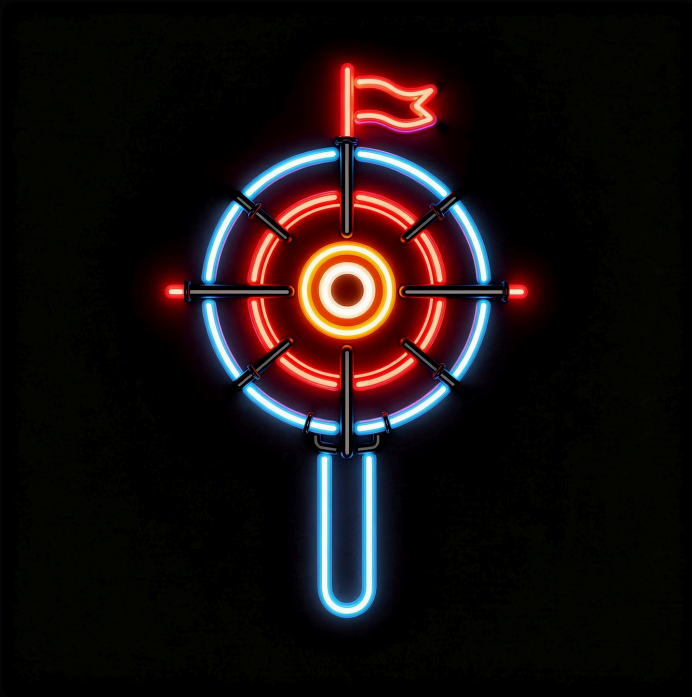
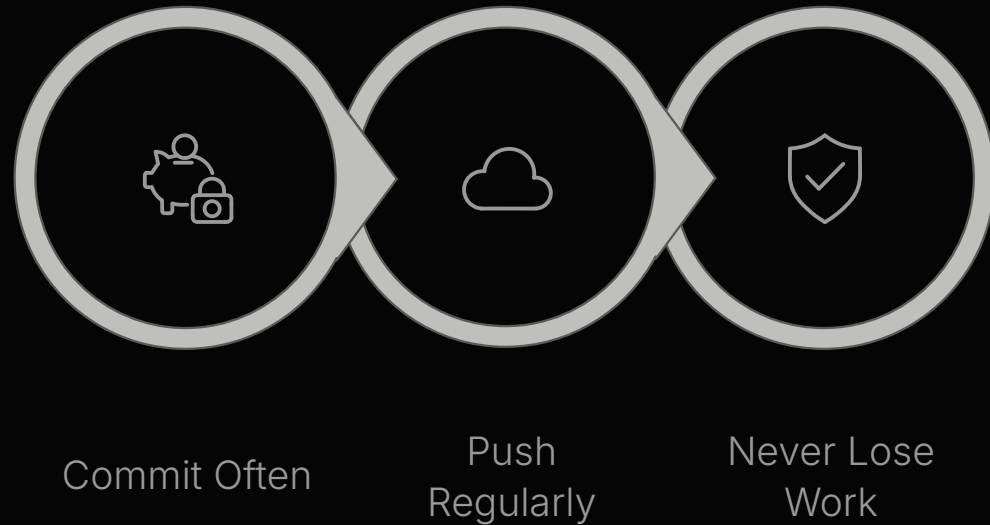
Runs commands, installs packages, starts servers

Git: Save Points for Your Code

Git = the save system. GitHub = the cloud backup.

- A **commit** is a snapshot — you can rewind anytime
- A **branch** is your own save file — no one else can break it
- Today everyone works on **firstname-lastname** branch

The rhythm:



5 Rules of the Road

1 Default to Sonnet, not Opus

Cheaper, faster, handles ~95% of the work just as well.

2 Fresh chat per feature

Long chats get expensive and confusing. Start clean.

3 Paste errors exactly

Don't describe the error — copy and paste the whole thing.

4 One clear spec beats ten chatty prompts

Think first. Say it clearly once. Then let the AI work.

5 Watch the preview, not just the code

If it looks broken in the browser, it *is* broken. Trust your eyes.

Ship Something You Can Text
to Your Family

Build it. Break it. Fix
it. Ship it.

By tomorrow morning, you'll have a real URL, real code, and a real story to
tell. Let's go.



LET'S BUILD

